Predicting Diabetes from Health and Lifestyle Indicators

A Support Vector Machine Approach Using NHIS 2022 Data

Bruna Porto | DATA 5322: Statistical Machine Learning II

Introduction

Diabetes affects roughly **10% of US adults** and is strongly linked to modifiable lifestyle factors such as diet, physical activity, and weight. Identifying which factors most strongly predict diabetes status from population-level survey data can inform screening strategies and public health priorities.

This work uses **Support Vector Machine** classifiers to predict whether a respondent has ever been diagnosed with diabetes, based on demographic and self-reported behavioral data from the **2022 National Health Interview Survey** (NHIS), accessed via IPUMS Health Surveys.

Three Kernels Compared

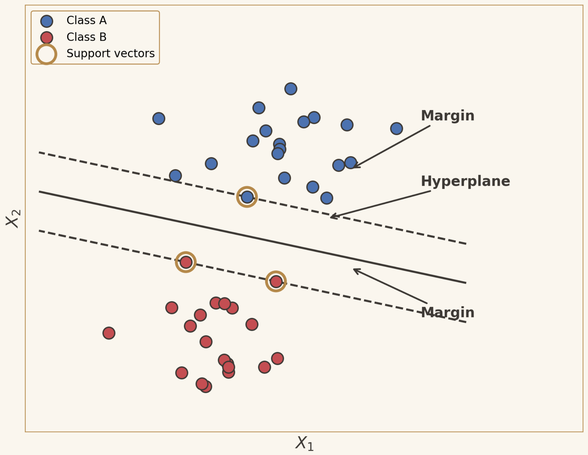
Linear	RBF	Polynomial
Flat decision boundary	Smooth nonlinear boundary	Curved boundary, fixed degree

Research Questions

1. Which predictors carry the most signal?
2. Does nonlinear flexibility improve over a linear baseline?
3. What do results suggest about prevention and screening?

Theoretical Background

SVMs find a decision boundary that separates two classes while **maximizing the margin** between the boundary and the closest training points (the *support vectors*). Only the support vectors determine the boundary.



Conceptual SVM in 2D: solid line = hyperplane, dashed = margins.

Key Considerations

- Features must be ***standardized***; SVMs are scale-sensitive.
- Tune C, γ, degree via ***cross-validation***; AUC preferred over accuracy when imbalanced.
- Note: scikit-learn's C inverts the textbook C — large sklearn C = strict margin.

Methodology

Data Preparation

NHIS 2022 subset to adults aged 18-85 (**n = 24,109** after cleaning). Special survey codes for unknown or out-of-universe responses were replaced with missing values per the codebook; complete cases retained. Target: ever diagnosed with diabetes (**10.4% prevalence**, 10:1 class imbalance).

Eight Predictors

Body composition
BMI

Demographics
Age, Sex

Physical activity
Moderate, vigorous (mins)

Lifestyle & SES
Sleep, soda, income ratio

Model Fitting Strategy

01 80/20 stratified train/test split

02 Standardization fit on training data only

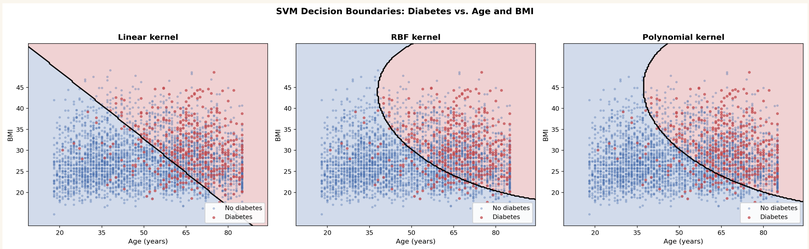
03 3-fold cross-validation, scoring = ROC-AUC

04 class_weight='balanced' for the 10:1 imbalance

Hyperparameter Grids

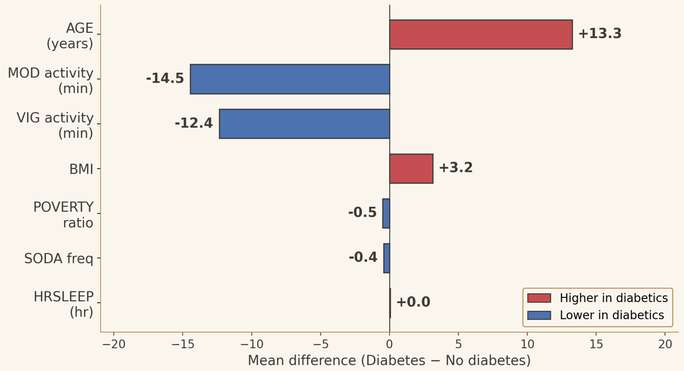
Kernel	C	Other parameters
Linear	0.1, 1, 10	—
RBF	0.5, 1, 2	γ ∈ {0.005, 0.01, 0.02}
Polynomial	0.1, 1, 10	deg ∈ {2, 3}, γ ∈ {scale, 0.01}

Decision Boundaries



Decision boundaries on AGE × BMI. The linear kernel cuts the space with a single line; RBF and polynomial kernels learn smooth curves capturing the joint AGE × BMI risk structure.

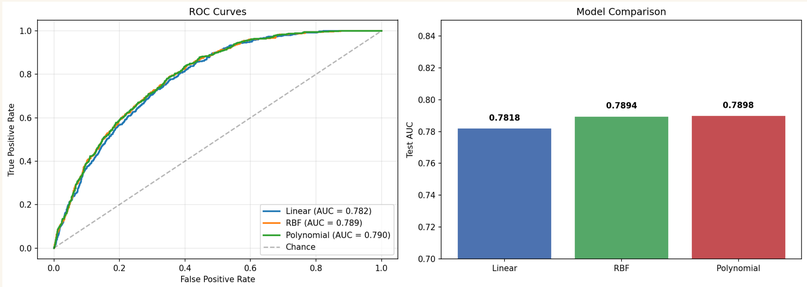
Predictor Mean Differences



Results

0.7818	0.7894	0.7898
Linear kernel	RBF kernel	Polynomial kernel
C = 1	C = 1, γ = 0.01	C = 1, deg = 3

Test-set performance (n = 4,822). All three kernels within 0.01 AUC of each other.



Discussion

Convergent kernels. All three reached test AUCs within 0.01 of each other (0.78-0.79). When models with very different inductive biases converge, the signal is well-captured by smooth, near-linear surfaces; the linear model is reasonable when interpretability matters.

Dominant predictors. Diabetic respondents were on average 13.3 years older with BMI 3.2 units higher and reported roughly half the physical activity. Age and BMI act as joint risk factors; each becomes sufficient at extremes.

Reverse causation in soda. Soda consumption was lower in diabetics, opposite of public-health predictions. This likely reflects post-diagnosis dietary changes; cross-sectional data captures association at one moment but cannot reveal directional effects.

Limitations

- Cross-sectional design prevents causal claims
- AGE topcoded at 85 compresses the upper risk tail
- Self-reported lifestyle variables introduce error
- Sample skews higher-income (median ~3.75× poverty)
- High recall (78-81%) but low precision (~20%) at 10:1

Conclusions

A high-recall, low-precision profile is appropriate for screening: missing a diabetic is more costly than a false alarm prompting follow-up.

BMI and physical activity are the strongest modifiable predictors at population scale; urban planning, subsidized recreation, and food-environment policy would be expected to reduce diabetes incidence. **Age-stratified screening** is most cost-effective for adults over 45.

References

[1] Blewett, L. A., Drew, J. A. R., King, M. L., et al. (2024). IPUMS Health Surveys: National Health Interview Survey, Version 7.4. Minneapolis, MN: IPUMS.

[2] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning, with Applications in Python (2nd ed.). Springer.

[3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830.

[4] Centers for Disease Control and Prevention (2024). National Diabetes Statistics Report.

Code: github.com/brunamportoDS/diabetes-svm-classification